

Khushi Piparava

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EDUCATION

Virginia Tech

Master of Science in Aerospace Engineering

Aug. 2026 – May 2028

University of Texas at Arlington

Honors Bachelor of Science in Aerospace Engineering, Minor in Mechanical Engineering

Aug. 2021 – Dec. 2025

Organizations: American Institute of Aeronautics and Astronautics (AIAA), Society of Asian Scientists and Engineers (SASE), Association for Computing Machinery (ACM)

SKILLS

CFD/FEA: Ansys (Fluent, Mechanical, Discovery), STAR-CCM+, SolidWorks (Static, Fluid, Thermal)

CAD & Mechanical Design: SolidWorks, Siemens NX, AutoCAD, CATIA

Aerodynamics & Flight Analysis: Lift/drag modeling, stability analysis, compressible flow, turbulence modeling

Aircraft Modeling & Analysis: OpenVSP, Digital DATCOM, Simulink, LabVIEW

Programming & Data: MATLAB, Python, C/C++, SQL, HTML/CSS

Machine Learning: Time-series modeling, predictive modeling, surrogate modeling for physical systems

Certifications: Machine Learning Specialization (Stanford & DeepLearning.AI)

EXPERIENCE

Research Assistant

Jan. 2026 – Present

Aerodynamics Research Center, UTA

Arlington, TX

- Developing ML-based time-series forecasting models to predict species evolution in methane pyrolysis experiments, training on **multi-condition datasets across 3 mixture fractions** to model hydrogen production.
- Built automated data augmentation pipelines for chemical kinetics datasets, generating **1000+ synthetic training points** and improving surrogate model prediction accuracy against experimental species time histories.

Software Testing Intern

Aug. 2024 – May 2025

Ansys (Synopsis)

Austin, TX

- Developed **50+ automated validation tests** for CAD geometry kernels, ensuring accurate part representation across multiple CAD formats like CATIA, UGNX, JT, and Creo, reducing regression errors and improving user reliability.
- Identified and resolved 30+ critical defects** introduced during the migration, collaborating with cross-functional teams to maintain compliance with product requirements and improve modeling stability.

Undergraduate Research Assistant

Aug. 2023 – Aug. 2025

Aerodynamics Research Center, UTA

Arlington, TX

- Conducted NASA-funded research on Rotating Detonation Engine nozzles using CFD and data-driven methods, **validating simulation results against experimental benchmarks with < 1% error.** (Co-authored publication)
- Developed reduced-order modeling (POD) workflows** for transient flowfields, reconstructing complex aerodynamic behavior using dominant modes and improving analysis speed by 30%. (First-author publication)

PROJECTS

Thermal R&D Design Tool - Heat Exchanger Modeling

Aug. 2025 – Dec. 2025

- Built a MATLAB-based thermal component model to size a forced-convection heat exchanger, solving for tube length **to meet a 20°C water temperature rise** using first-principles energy balance and convection correlations.
- Performed 3-variable parametric sweeps** (tube diameter, air velocity, and temperature) and generated design plots to identify feasible operating regions and thermal tradeoffs for rapid system optimization.

Conceptual Design of Supersonic Business Jet

Jun. 2025 – Aug. 2025

- Led Aerodynamics & Aerothermodynamics for a 9-person team generating **validated lift-drag datasets** using Digital DATCOM, OpenVSP, and MATLAB for a Mach 1.7, 4,500 nmi, 14-passenger tailless SBJ on 100% SAF.
- Analyzed static stability, confirming **> 8% static margin** and safe control response across the full Mach range, supporting multidisciplinary vehicle design and performance evaluation.

Formula SAE Aerodynamic Analysis Project

Jun. 2024 – Aug. 2024

- Redesigned assembly of UTA Racing's A-modified car in SolidWorks and simulated aerodynamic performance using STAR CCM+ CFD analysis, achieving a **12% increase in downforce**.
- Conducted ride-height and wing-angle parametric studies, identifying configurations that improved **aerodynamic balance by 10%** for enhanced vehicle stability.